Multinomial Pascal's Triangle

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Abstract

We try to arrange the terms of a multinomial expansion in $\mathbb{R}^n$, as a generalization of the well-known Pascal's Triangle. Starting from a few constraints suggested by the ordinary triangle, we show that such an arrangement exists. We then show that all lattice arrangements are related by orthogonal matrices: fixing one of them gives a bijection between the set of lattice Multinomial Pascal's Triangles and the set of orthogonal matrices. Finally, we show that even without assuming a lattice structure, the initial vectors uniquely determine the arrangement.

1 Introduction

$$\begin{aligned}
(a+b)^0 &= \mathbf{1} \\
(a+b)^1 &= \mathbf{1}a + \mathbf{1}b \\
(a+b)^2 &= \mathbf{1}a^2 + \mathbf{2}ab + \mathbf{1}b^2 \\
(a+b)^3 &= \mathbf{1}a^3 + \mathbf{3}a^2b + \mathbf{3}ab^2 + \mathbf{1}b^3 \\
(a+b)^4 &= \mathbf{1}a^4 + \mathbf{4}a^3b + \mathbf{6}a^2b^2 + \mathbf{4}ab^3 + \mathbf{1}b^4
\end{aligned}$$

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

Pascal's Triangle is a clever arrangement of the terms of binomial expansions. The kth row of the triangle contains the coefficients of the expansion of $(a+b)^k$. Usually the variables a, b are left out for simplicity. The triangle has two basic, interesting properties.

1. The arrangement is visually appealing.
2. Each binomial coefficient is the sum of the two coefficients directly above it.

I wondered whether a similar picture could be made for the coefficients of multinomial expansions. The properties above, specifically "visually appealing" and "above", are not well defined. Let's try to define them, so that we can use these definitions as constraints for a Multinomial Pascal's Triangle. I will try to arrange the multinomial terms in $\mathbb{R}^n$.

Definition 1.1. *An **n-nomial Pascal's Triangle** is an arrangement of the terms of* $(a_1 + a_2 + \cdots + a_n)^k$*, for all* $k \in \mathbb{N} \cup \{0\}$*, in some* $\mathbb{R}^m$*.*

We will use the words multinomial and n-nomial interchangeably, using n-nomial when n has to be specified.

2 Defining Constraints

2.1 What does "visually appealing" mean?

"Visually appealing" is ambiguous at best. One could say that the key quality is that the binomial terms form an equilateral triangle, but that does not generalize well to higher dimensions. Here are some more basic observations.

1. Each term is the same distance away from its "adjacent" terms.
2. The terms sit on a grid-like structure.

Now let's turn these observations into constraints.

Bound 1. *In the triangle above, $2ab$ is the same distance from $1a$, $1b$, $1a^2$, $1b^2$, $3a^2b$, and $3ab^2$. Each of these is obtained from ab by adding one variable (e.g. $ab \to a^2b$), removing one variable (e.g. $ab \to b$), or swapping one variable for another (e.g. $ab \to a^2$). We call two terms* ***adjacent*** *if one can be turned into the other by exactly one such addition, removal, or swap, and we require every pair of adjacent terms to be the same distance apart.*

Bound 2. *Let the term a^0b^0 sit at the origin, and let $\vec{a}, \vec{b}$ be the vectors from the origin to the terms a and b. Then the term a^kb^l sits at $k\vec{a} + l\vec{b}$, so the terms form a lattice.*

Bound 2 is a strong constraint: the entire arrangement is determined by the initial vectors.

Definition 2.1. *A Multinomial Pascal's Triangle that satisfies Bound 2 is called a* ***Lattice Multinomial Pascal's Triangle****.*

2.2 What do "sum" and "above" mean?

The numerical property of Pascal's Triangle is that each coefficient equals the sum of the two coefficients above it. Before going further, let's define some terms.

2.2.1 Terms

In Pascal's Triangle, the terms of $(a+b)^k$ form a line, and the terms of $(a+b)^k$ sit "above" the terms of $(a+b)^{k+1}$.

Definition 2.2. *The k-**layer** of an n-nomial Pascal's Triangle is the set of locations of the terms of $(a_1 + a_2 + \cdots + a_n)^k$.*

From this observation, we can require each layer to lie on a hyperplane, with all of these hyperplanes parallel. "Aboveness" is then just the position along the line orthogonal to the hyperplanes.

Bound 3. *Suppose an n-nomial Pascal's Triangle is arranged in $\mathbb{R}^m$. Each k-layer lies on a hyperplane H_k of $\mathbb{R}^m$. All of these hyperplanes are parallel to each other, and different layers lie on different hyperplanes.*

Remark 1. *The 0-layer is a single point, so it lies on many hyperplanes. We take H_0 to be the one through that point that is parallel to the others.*

Remark 2. *The last part of Bound 3 ("different layers lie on different hyperplanes") is important. Without it, a layer is allowed to fold back onto the hyperplane of an earlier layer, and the uniqueness result in Section 5 fails.*

As a final constraint, we want to arrange Pascal's Triangle in the lowest dimension possible.

Bound 4. *If an arrangement of n-nomial terms satisfying the bounds is possible in $\mathbb{R}^k$, then an arrangement that requires $\mathbb{R}^p$ for $p > k$ is not considered an n-nomial Pascal's Triangle.*

2.2.2 Recovering the sum property

In this subsection, we show that if an arrangement satisfies the bounds above, the property "each coefficient is the sum of the coefficients above it" still holds, and the summands sit in a nice position.

Definition 2.3. *For a term $t = a_1^{x_1}a_2^{x_2}\cdots a_n^{x_n}$, the terms* ***directly above*** *t are the terms $\{t/a_i \mid 1 \le i \le n,\ x_i \ge 1\}$.*

Theorem 2.1.

$$\binom{n}{a_1, a_2, \ldots, a_k} = \sum_{i=1}^{k} \binom{n-1}{a_1, \ldots, a_i - 1, \ldots, a_k}$$

Proof. We can prove this by induction, but there is a more intuitive proof. $\binom{n}{a_1,a_2,\ldots,a_k}$ is the number of ways to divide n elements into k groups, with a_i elements in the ith group. Pick the first element, and split into cases by which group it goes to. If it goes to the ith group, we are left with $n-1$ elements to divide into k groups, with $a_i - 1$ elements in the ith group. There are $\binom{n-1}{a_1,\ldots,a_i-1,\ldots,a_k}$ ways to do this. Summing over i gives the result. □

So the coefficients of the terms directly above t add up to the coefficient of t.

Remark 3. *For any term t, the terms directly above t are adjacent to t (they differ by removing one variable) and lie in the layer above t.*

So with these constraints, the sum property of the coefficients holds, and the summands are nicely located: they are exactly the neighbors of t in the layer above.

3 Existence of a Multinomial Pascal's Triangle

Let's now show that there is an arrangement of multinomial terms that satisfies the above requirements.

First, look at the zeroth and first layers. Let $\vec{a_i}$ be the vector from the origin to the term a_i. Since $||\vec{a_i}||$ must be the same for all i, let's set $||\vec{a_i}|| = 1$.

Also, for any $i \neq j$, Bound 1 requires

$$||a_i|| = ||a_j|| = ||a_i - a_j||.$$

By the law of cosines,

$$\begin{aligned} ||a_i - a_j||^2 &= ||a_i||^2 + ||a_j||^2 - 2||a_i||\,||a_j|| \cos\theta_{ij} \\ 1 &= 1 + 1 - 2\, a_i \cdot a_j \\ a_i \cdot a_j &= \frac{1}{2} \end{aligned}$$

Let's now show that such vectors exist in $\mathbb{R}^n$. We will choose them so that a_j has nonzero entries only in its first j coordinates.

Set $a_1 = [1, 0, 0, \cdots, 0]$. Since $a_1 \cdot a_2 = \frac{1}{2}$, the first entry of a_2 must be $\frac{1}{2}$, and then $||a_2|| = 1$ forces the second entry to be $\pm\frac{\sqrt{3}}{2}$. Let's pick $a_2 = [\frac{1}{2}, \frac{\sqrt{3}}{2}, 0, \cdots, 0]$.

Let's do one more. For a_3, the dot products with a_1 and a_2 fix the first two entries: $a_3 = [\frac{1}{2}, \frac{1}{\sqrt{12}}, \cdots]$. The third entry is then fixed (up to sign) by $||a_3|| = 1$.

We can keep going like this to create each new a_i. Let's prove this formally by induction.

Theorem 3.1. *For any $n \in \mathbb{N}$, there exist vectors $a_1, \ldots, a_n \in \mathbb{R}^n$ such that*

$$a_i \cdot a_j = \frac{1}{2} \text{ for all } i \neq j, \qquad ||a_i|| = 1 \text{ for all } i.$$

Proof. Proof by induction. For the base case, set $a_1 = [1, 0, \cdots, 0]$.

Now suppose we have $a_1, \ldots, a_k$, where $a_k = [a_k^1, a_k^2, \cdots, a_k^k, 0, \cdots, 0]$ with $a_k^k \neq 0$. We construct a_{k+1}. Its first $k-1$ entries must equal those of a_k, since both vectors have the same dot products with $a_1, \ldots, a_{k-1}$. The kth entry is fixed by $a_k \cdot a_{k+1} = \frac{1}{2}$, and the $(k+1)$th entry by $||a_{k+1}|| = 1$:

$$a_{k+1} = \left[a_k^1, \cdots, a_k^{k-1},\ \frac{\frac{1}{2} - \sum_{i=1}^{k-1}(a_k^i)^2}{a_k^k},\ \pm\sqrt{1 - \sum_{i=1}^{k}(a_{k+1}^i)^2},\ 0, \cdots, 0 \right].$$

It remains to check that the number under the square root is positive. One way to see this is that the Gram matrix of the a_i (the matrix with 1 on the diagonal and $\frac{1}{2}$ everywhere else) has eigenvalues $\frac{1}{2}$ and $\frac{n+1}{2}$, so it is positive definite. This means every step of the construction succeeds. □

Corollary 3.2. *In $\mathbb{R}^n$, if $a_1, a_2, \cdots, a_{n-1}$ are fixed, there are only two choices for a_n.*

We have found a set of vectors that satisfies the equal-distance constraint for layers 0 and 1. In fact, using these vectors in the grid-like (lattice) arrangement gives an n-nomial Pascal's Triangle in $\mathbb{R}^n$.

In the lattice arrangement, the distance between two adjacent terms is always one of

$$||\vec{a_i}||, \qquad ||\vec{a_i} - \vec{a_j}||,$$

and these are all equal to 1 by the constraint on the first two layers.

Before checking Bound 3, let's prove a simple observation.

Lemma 3.3. *Any set of vectors a_i that satisfies the Multinomial Pascal's Triangle constraints must form a basis.*

Proof. For the sake of contradiction, suppose the a_i do not form a basis. Then, without loss of generality, there are constants c_i with

$$\sum_{i=1}^{n-1} c_i a_i = a_n.$$

Taking the dot product with a_j, for any $j \leq n-1$:

$$\begin{aligned} a_n \cdot a_j &= \frac{1}{2} \\ \Big(\sum_{i=1}^{n-1} c_i a_i\Big) \cdot a_j &= \frac{1}{2} \\ \frac{1}{2} c_j + \frac{1}{2} \sum_{i=1}^{n-1} c_i &= \frac{1}{2} \\ c_j + \sum_{i=1}^{n-1} c_i &= 1 \end{aligned}$$

Taking the dot product with a_n instead:

$$\begin{aligned} a_n \cdot a_n &= 1 \\ a_n \cdot \sum_{i=1}^{n-1} c_i a_i &= 1 \\ \frac{1}{2} \sum_{i=1}^{n-1} c_i &= 1 \\ \sum_{i=1}^{n-1} c_i &= 2 \end{aligned}$$

So $c_j = 1 - 2 = -1$ for every j from 1 to $n-1$. But then $\sum_{i=1}^{n-1} c_i = -(n-1) \neq 2$, which is a contradiction. □

Now let's check Bound 3. The k-layer consists of the points $\sum_i x_i a_i$ with $\sum_i x_i = k$. We can rewrite these as $ka_1 + \sum_{i \geq 2} x_i(a_i - a_1)$. Since the a_i are linearly independent, the vectors $a_i - a_1$ $(i \geq 2)$ span an $(n-1)$-dimensional subspace, so the k-layer lies on the hyperplane $ka_1 + \text{span}\{a_i - a_1\}$. These hyperplanes are parallel, and they are different for different k.

So there is a Multinomial Pascal's Triangle that satisfies the constraints.

Theorem 3.4. *An n-nomial Pascal's Triangle must lie in $\mathbb{R}^n$.*

Proof. By Lemma 3.3, at least n dimensions are needed. We also found an n-nomial Pascal's Triangle in $\mathbb{R}^n$. So by Bound 4, every n-nomial Pascal's Triangle must lie in $\mathbb{R}^n$. □

From now on, Bound 4 simply means that an n-nomial Pascal's Triangle lies in $\mathbb{R}^n$.

4 Generating All Lattice Multinomial Pascal's Triangles

In the last section, we showed that a lattice Multinomial Pascal's Triangle exists. In this section, we find all of them.

Lemma 4.1. *If $\{v_1, v_2, \cdots, v_n\}$ and $\{w_1, w_2, \cdots, w_n\}$ are both bases of $\mathbb{R}^n$, then there is a matrix W such that $W[v_1, v_2, \cdots, v_n] = [w_1, w_2, \cdots, w_n]$.*

Proof. This is the change of basis theorem from linear algebra. □

Lemma 4.2. *Given a basis X of a Lattice Multinomial Pascal's Triangle and an orthogonal matrix A, AX is also a basis of a Lattice Multinomial Pascal's Triangle.*

Proof. An orthogonal matrix preserves lengths and angles. So the new vectors AX also satisfy the bounds. □

Now let $V = [v_1, v_2, \cdots, v_n]$ and $W = [w_1, w_2, \cdots, w_n]$ be the initial vectors (as columns) of two Lattice Multinomial Pascal's Triangles. Both sets of vectors have the same lengths and the same pairwise dot products, so

$$V^T V = W^T W.$$

By Lemma 4.1, there is a change of basis matrix $A = WV^{-1}$. Then

$$A^T A = V^{-T} W^T W V^{-1} = V^{-T} V^T V V^{-1} = I.$$

So A is an orthogonal matrix.

Lemma 4.3. *For any two bases V, W of Lattice Multinomial Pascal's Triangles, the change of basis matrix A with $AV = W$ is an orthogonal matrix.*

Proof. See the computation above. □

Theorem 4.4. *Applying all orthogonal matrices to the basis of one Lattice n-nomial Pascal's Triangle generates every Lattice n-nomial Pascal's Triangle.*

Proof. By Lemma 4.1, there is a change of basis matrix between any two bases of Lattice n-nomial Pascal's Triangles, and by Lemma 4.3 that matrix is orthogonal. Together with Lemma 4.2, this shows that the orbit of one basis under the orthogonal matrices is exactly the set of all bases. □

Theorem 4.5. *Fix one Lattice n-nomial Pascal's Triangle with basis X_0. Then $A \mapsto AX_0$ is a bijection between the set of orthogonal matrices and the set of bases of Lattice n-nomial Pascal's Triangles.*

Proof. By Theorem 4.4, the map is surjective.

The map is also injective: if $AX_0 = BX_0$, then $A = B$ because X_0 is invertible. □

5 Loosening the Bounds

5.1 Uniqueness of the binomial Pascal's Triangle

We have found all the triangles that satisfy the bounds. What happens if we drop the lattice-structure bound (Bound 2)?

Then there seem to be several possible binomial Pascal's Triangles. For example, we could build layer 2 under layer 1 as usual,

$$\begin{array}{ccc} & 1a & 1b \\ 1a^2 & 2ab & 1b^2 \end{array}$$

or flip it upward so that it lands on the same line as the top term 1:

$$\begin{array}{ccc} 1a^2 & 2ab & 1b^2 \\ 1a & & 1b \end{array}$$

But the flipped version puts layer 2 on the same line as layer 0, which Bound 3 does not allow. In this section, we show that once the first layer is fixed, there is exactly one binomial Pascal's Triangle satisfying Bounds 1, 3, and 4.

Definition 5.1. *The **initial vectors** of an n-nomial Pascal's Triangle are the vectors from the origin to the terms of the first layer.*

By Bound 1 and Lemma 3.3, these vectors form a basis of $\mathbb{R}^n$.

Theorem 5.1. *Given the initial vectors, there is exactly one binomial Pascal's Triangle.*

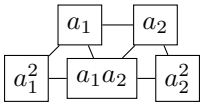
Proof. Layers 0 and 1 are fixed by the initial vectors.

Let's look at layers 1 and 2. Layer 2 has three terms and lies on a line. There are two possible configurations for layer 2:

$$a_1^2 - -a_1a_2 - -a_2^2$$
$$(a_1^2, a_2^2) - -a_1a_2$$

In the second one, a_1^2 and a_2^2 sit at the same point. We will see below why that is impossible. Let's start with the first configuration.

Attaching layer 2 to layer 1, we get the following picture.



By the adjacency constraint, every triangle formed by adjacent terms is equilateral, so every angle in it is 60 degrees. Also, a_1^2, a_1a_2, a_2^2 lie on a line by Bound 3. Writing $\overrightarrow{p, q}$ for the vector from p to q, we get

$$\begin{aligned}(\overrightarrow{a_1^2, a_1a_2} - \overrightarrow{a_1^2, a_1}) \cdot \overrightarrow{a_1, a_2} &= \frac{1}{2}\\ \overrightarrow{a_1^2, a_1a_2} \cdot \overrightarrow{a_1, a_2} - \overrightarrow{a_1^2, a_1} \cdot \overrightarrow{a_1, a_2} &= \frac{1}{2}\\ \overrightarrow{a_1^2, a_1a_2} \cdot \overrightarrow{a_1, a_2} &= 1\end{aligned}$$

So $\overrightarrow{a_1^2, a_1a_2}$ and $\overrightarrow{a_1, a_2}$ are parallel.

This also means that $\angle(a_1^2, a_1a_2, a_2) = 120$ degrees. If a_1^2 and a_2^2 were at the same point, then $|\overline{a_2\, a_2^2}| > 1$. So the second configuration is impossible.

So the angle between $\overrightarrow{a_1^2, a_1}$ and $\overrightarrow{a_1, a_1a_2}$ is 60 degrees. In $\mathbb{R}^2$, this leaves only two possible values for $\overrightarrow{a_1^2, a_1}$: one is $\overrightarrow{a_1, 0}$, and the other is its reflection across the line $\overline{a_1a_2}$. The same argument works for $\overrightarrow{a_2^2, a_2}$.

Since a_1^2, a_1a_2, a_2^2 must lie on a line with a_1^2 and a_2^2 at distance 2, the vectors $\overrightarrow{a_1^2, a_1}$ and $\overrightarrow{a_2^2, a_2}$ must either both point towards the origin or both point away from it. If both point away from the origin, layer 2 ends up on the same line as layer 0, which Bound 3 does not allow.

The same argument works for adding each further layer: at every step, there is only one way to attach the new layer to the triangle. (We prove this carefully, for every n, in the next subsection.)

So there is only one binomial Pascal's Triangle. □

5.2 Uniqueness of the n-nomial Pascal's Triangle

In this section, we show that even without Bound 2, every n-nomial Pascal's Triangle is a Lattice n-nomial Pascal's Triangle. The idea is the same as in the binomial case: we attach one layer at a time, and show that at each step there is only one way to attach it.

By Bound 4, the triangle lives in $\mathbb{R}^n$, so by Lemma 3.3 the initial vectors $a_1, \ldots, a_n$ form a basis of $\mathbb{R}^n$.

Let's first set up some notation for the layers. By Bound 3, all the layers lie on parallel hyperplanes, so they share a unit normal vector w. Every a_i lies on H_1, so $w \cdot a_i$ is the same number for all i. Let's call it h. Since $H_1 \neq H_0$, $h \neq 0$, and we can choose the direction of w so that $h > 0$.

For a lattice, $w \cdot (x_1a_1 + \cdots + x_na_n) = (x_1 + \cdots + x_n)h$. So in a lattice, the k-layer lies on the hyperplane $\{v : w \cdot v = kh\}$. In other words, the layers are evenly spaced, h apart.

Lemma 5.2. *$a_1 + a_2 + \cdots + a_n$ is parallel to w.*

Proof. Let $s = a_1 + \cdots + a_n$. For every l, $s \cdot a_l = 1 + \frac{n-1}{2} = \frac{n+1}{2}$, the same number for all l. Since $w \cdot a_l = h$ for all l as well, the vector $s - \frac{n+1}{2h}w$ has dot product 0 with every a_l. The a_l form a basis, so this vector is 0. □

Now let's look at the triangle from the point of view of one term. Pick any term t in the $(k-2)$-layer, and any i. The term ta_i is in the $(k-1)$-layer, and the terms $ta_ia_1, ta_ia_2, \cdots, ta_ia_n$ are in the k-layer (one of them is ta_i^2). Let

$$v_j = \overrightarrow{ta_i,\, ta_ia_j}, \qquad j = 1, \ldots, n.$$

Each ta_ia_j is adjacent to ta_i (add a_j), so $||v_j|| = 1$. Also, ta_ia_j and ta_ia_l are adjacent (swap a_j for a_l), so $||v_j - v_l|| = 1$, and as in Section 3 this gives $v_j \cdot v_l = \frac{1}{2}$. So the vectors v_j satisfy the same condition as the initial vectors.

By the computation in Section 4 (just before Lemma 4.3), this means there is an orthogonal matrix Q_i with $v_j = Q_ia_j$ for all j. In words: the way the k-layer is attached below ta_i is a rotated (or reflected) copy of the way the first layer is attached below the origin. The lattice is the case where every Q_i is the identity. We will show that this is the only possibility.

Theorem 5.3. *Suppose an n-nomial Pascal's Triangle is a lattice up to the $(k-1)$-layer. Then for every term t in the $(k-2)$-layer and every i, j,*

$$\overrightarrow{ta_i,\, ta_ia_j} = a_j.$$

Proof. Fix t, and let $Q_1, \ldots, Q_n$ be the orthogonal matrices above. The term ta_ia_j can be reached from t by first going to ta_i, so

$$\overrightarrow{t,\, ta_ia_j} = a_i + Q_ia_j.$$

Step 1: only two choices, and one of them folds back. All terms of the k-layer lie on H_k, so $w \cdot \overrightarrow{t,\, ta_ia_j} = h + w \cdot Q_ia_j$ is the same for all j. Let $u = Q_i^T w$. Then $w \cdot Q_ia_j = u \cdot a_j$, so $u \cdot a_j$ is the same number d for all j. We also have $w \cdot a_j = h$ for all j, so, as in the proof of Lemma 5.2, $u = \frac{d}{h}w$. Since Q_i is orthogonal, $||u|| = ||w|| = 1$, so $d = h$ or $d = -h$.

So, just like in the binomial case, there are only two choices. If $d = -h$, then $w \cdot \overrightarrow{t,\, ta_ia_j} = 0$, so the k-layer lands on the hyperplane of the $(k-2)$-layer, which Bound 3 does not allow. So $d = h$ and $u = w$, which means $Q_iw = w$. In words: Q_i cannot tilt the layer. It can only rotate it within the hyperplane.

Step 2: no rotation. The term ta_ia_j is below both ta_i and ta_j, so we can reach it from t in two ways:

$$a_i + Q_ia_j = a_j + Q_ja_i.$$

Let's sum this over j, and write $s = a_1 + \cdots + a_n$:

$$na_i + Q_is = s + (Q_1 + \cdots + Q_n)a_i.$$

By Lemma 5.2, s is parallel to w, and by Step 1, $Q_iw = w$. So $Q_is = s$, and we are left with

$$(Q_1 + \cdots + Q_n)a_i = na_i.$$

This holds for every i, and the a_i form a basis, so the average $\frac{1}{n}(Q_1 + \cdots + Q_n)$ is the identity.

But the average of orthogonal matrices can only be the identity if every one of them is the identity. To see this, take any unit vector v. The vectors $Q_1v, \ldots, Q_nv$ are unit vectors whose average is the unit vector v, and an average of unit vectors only has length 1 if they are all the same vector. So $Q_jv = v$ for every v, i.e. $Q_j = I$.

So $\overrightarrow{ta_i,\, ta_ia_j} = Q_ia_j = a_j$. □

Theorem 5.4. *The initial vectors uniquely determine the n-nomial Pascal's Triangle.*

Proof. Every Multinomial Pascal's Triangle is a lattice triangle up to layer 1. Now suppose it is a lattice up to the $(k-1)$-layer, for some $k \geq 2$. Every term of the k-layer can be written as ta_ia_j for some term t in the $(k-2)$-layer. By Theorem 5.3, the location of ta_ia_j is the location of t plus $a_i + a_j$, which is exactly where the lattice puts it. So the triangle is a lattice up to the k-layer. By induction, every Multinomial Pascal's Triangle is a Lattice Multinomial Pascal's Triangle. This means the initial vectors uniquely determine the Multinomial Pascal's Triangle. □

Remark 4. *The proof never assumes that different terms sit at different points. That comes for free.*